14.3: Inheritance patterns are often more complex than predicted by simple Mendelian genetics

In the 1900s, geneticists extended Mendelian principles not only to diverse organisms, but also to patterns of inheritance more complex than those described by Mendel. For the work that led to his two laws of inheritance, Mendel chose pea plant characters that turn out to have a relatively simple genetic basis: Each character is determined by one gene, for which there are only two alleles, one completely dominant and the other completely recessive. (There is one exception: Mendel's pod shape character is actually determined by two genes.) Not all heritable characters are determined so simply, and the relationship between genotype and phenotype is rarely so straightforward. Mendel himself realized that he could not explain the more complicated patterns he observed in crosses involving other pea characters or other plant species. This does not diminish the utility of Mendelian genetics, however, because the basic principles of segregation and independent assortment apply even to more complex patterns of inheritance. In this section, we will extend Mendelian genetics to hereditary patterns that were not reported by Mendel.

Extending Mendelian Genetics for a Single Gene

The inheritance of characters determined by a single gene deviates from simple Mendelian patterns when alleles are not completely dominant or recessive, when a particular gene has more than two alleles, or when a single gene produces multiple phenotypes. We will describe examples of each of these situations in this section.

Degrees of Dominance

Alleles can show different degrees of dominance and recessiveness in relation to each other. In Mendel's classic pea crosses, the F1 offspring always looked like one of the two parental varieties because one allele in a pair showed **complete dominance** over the other. In such situations, the phenotypes of the heterozygote and the dominant homozygote are indistinguishable (see Figure 14.6).

For some genes, however, neither allele is completely dominant, and the F1 hybrids have a phenotype somewhere between those of the two parental varieties. This phenomenon, called **incomplete dominance**, is seen when red snapdragons are crossed with white snapdragons: All the F1 hybrids have pink flowers **(Figure 14.10)**. This third, intermediate phenotype results from flowers of the heterozygotes having less red pigment than the red homozygotes. (This is unlike the case of Mendel's pea plants, where the

Pp heterozygotes make enough pigment for the flowers to be purple, indistinguishable from those of PP plants.)

Figure 14.10

```
Incomplete dominance in snapdragon color.
```

P Generation = Red flower crossed with white flower. F 1 Generation = all pink flowers. F 2 Generation = 1 Red to 2 pink to 1 white flowers.

Gametes	one-half C superscript R	one-half C superscript W
one-half C superscript R	C superscript R, C superscript R. Phenotype: Red flower	C superscript R, C superscript W. Phenotype: Pink flower
one-half C superscript W	C superscript R, C superscript W. Phenotype: Pink flower	C superscript W, C superscript W. Phenotype: Pink flower

P Generation: Red Flower (C superscript R, C superscript R) x White Flower (C superscript W, C superscript W). Arrows indicate red flower gamete (C superscript R) crossed with white flower gamete (C superscript W), flowing to F 1 Generation.

F 1 Generation: All pink flowers (C superscript W, C superscript W). Arrows indicate gametes are one-half C superscript R and one-half C superscript W. F 2 Generation: Punnett square with eggs on the left side and sperm on top. A Punnett square is represented by the following table.

Gametes	one-half C superscript R	one-half C superscript W
one-half C superscript R	C superscript R, C superscript R. Phenotype: Red flower	C superscript R, C superscript W. Phenotype: Pink flower
one-half C superscript W	C superscript R, C superscript W. Phenotype: Pink flower	C superscript W, C superscript W. Phenotype: Pink flower

When red snapdragons are crossed with white ones, the F1 hybrids have pink flowers. Segregation of alleles into gametes of the F1 plants results in an F2 generation with a 1:2:1 ratio for both genotype and phenotype. Neither allele is dominant, so rather than using upper- and lowercase letters, we use the letter *C* with a superscript to indicate an allele for flower color: CR for red and CW for white.

Suppose a classmate argues that this figure supports the blending hypothesis for inheritance. What might your classmate say, and how would you respond?

*For suggested answer, see*Appendix A.**

Animation

Incomplete Dominance in “MendAliens”

image

Play

image

Play

At first glance, incomplete dominance of either allele seems to provide evidence for the blending hypothesis of inheritance, which would predict that the red or white trait could never reappear among offspring of the pink hybrids. In fact, interbreeding F1 hybrids produces F2 offspring with a phenotypic ratio of one red to two pink to one white. (Because heterozygotes have a separate phenotype from homozygotes, the genotypic and phenotypic ratios for the F2 generation are the same, 1:2:1.) The segregation of the red-flower and white-flower alleles in the gametes produced by the pink-flowered plants confirms that the alleles for flower color are heritable factors that maintain their identity in the hybrids; that is, that the factors are discrete rather than “blendable.”

Another variation on dominance relationships between alleles is called **codominance**; in this variation, the two alleles each affect the phenotype in separate, distinguishable ways. For example, the human MN blood group is determined by codominant alleles for two specific molecules located on the surface of red blood cells, the M and N molecules. A single gene (L), for which two allelic variations are possible (LM or LN), determines the phenotype of this blood group. Individuals homozygous for the LM allele LMLM have red blood cells with only M molecules; individuals homozygous for the LN allele LNLN have red blood cells with only N molecules. But *both* M and N molecules are present on the red blood cells of individuals heterozygous for the M and N alleles LMLN. Note that the MN phenotype is *not* intermediate between the M and N phenotypes, which distinguishes codominance from incomplete dominance. Rather, *both* M and N phenotypes are exhibited by heterozygotes, since both molecules are present.

The Relationship Between Dominance and Phenotype

We’ve now seen that the relative effects of two alleles range from complete dominance of one allele to incomplete dominance of either allele to codominance of both alleles. It is important to understand that an allele is called *dominant* because it is seen in the phenotype, not because it somehow subdues a recessive allele. Alleles are simply variations in a gene’s nucleotide sequence (see Figure 14.4). When a dominant allele

coexists with a recessive allele in a heterozygote, they do not actually interact at all. It is in the pathway from genotype to phenotype that dominance and recessiveness come into play.

To illustrate the relationship between dominance and phenotype, we can use one of the characters Mendel studied—round versus wrinkled pea seed shape. The dominant allele (round) codes for an enzyme that helps convert an unbranched form of starch to a branched form in the seed. The recessive allele (wrinkled) codes for a defective form of this enzyme, leading to an accumulation of unbranched starch, which causes excess water to enter the seed by osmosis. Later, when the seed dries, it wrinkles. If a dominant allele is present, no excess water enters the seed and it does not wrinkle when it dries. One dominant allele results in enough of the enzyme to synthesize adequate amounts of branched starch, which means that dominant homozygotes and heterozygotes have the same phenotype: round seeds.

A closer look at the relationship between dominance and phenotype reveals an intriguing fact: For any character, the observed dominant/recessive relationship of alleles depends on the level at which we examine phenotype. **Tay-Sachs disease**, an inherited disorder in humans, is an example. The brain cells of a child with Tay-Sachs disease cannot metabolize certain lipids because a crucial enzyme does not work properly. As these lipids accumulate in brain cells, the child begins to experience seizures, blindness, and degeneration of motor and mental performance and dies within a few years.

Only children who inherit two copies of the Tay-Sachs allele (homozygotes) have the disease. Thus, at the *organismal* level, the Tay-Sachs allele qualifies as recessive. However, the activity level of the lipid-metabolizing enzyme in heterozygotes is intermediate between the activity level in individuals homozygous for the normal allele and the activity level in individuals with Tay-Sachs disease. (The term *normal* is used in the genetic sense to refer to the allele coding for the enzyme that functions properly.) The intermediate phenotype observed at the *biochemical* level is characteristic of incomplete dominance of either allele. Fortunately, the heterozygote condition does not lead to disease symptoms, apparently because half the normal enzyme activity is sufficient to prevent lipid accumulation in the brain. Extending our analysis to yet another level, we find that heterozygous individuals produce equal numbers of normal and dysfunctional enzyme molecules. Thus, at the *molecular* level, the normal allele and the Tay-Sachs allele are codominant. As you can see, whether alleles appear to be completely dominant, incompletely dominant, or codominant depends on the level at which the phenotype is analyzed.

Frequency of Dominant Alleles

While you might assume that the dominant allele for a particular character would be more common than the recessive allele, this is not always the case. For an example of a rare dominant allele, about one baby out of 400 in the United States is born with extra fingers or toes, a condition known as polydactyly. Some cases are caused by the presence of a dominant allele. The low frequency of these types of polydactyly indicates that

the recessive allele, which results in five digits per appendage, is far more prevalent than the dominant allele in the population.

Multiple Alleles

Only two alleles exist for the pea characters that Mendel studied, but most genes have more than two alleles. The ABO blood groups in humans, for instance, are determined by the two alleles a person has of the blood group gene; the three possible alleles are IA, IB, and i. A person's blood group may be one of four types: A, B, AB, or O. These letters refer to two carbohydrates—A and B—that may be found attached to specific cell-surface molecules on red blood cells. An individual's blood cells may have carbohydrate A (type A blood), carbohydrate B (type B), both (type AB), or neither (type O), as shown in **Figure 14.11**, along with the relevant genotypes. Matching compatible blood groups is critical for safe blood transfusions (see Concept 43.3).

Figure 14.11

Multiple alleles for the ABO blood groups.

Multiple alleles for the A B O blood groups.

The four blood groups result from different combinations of three alleles.

Allele	upper I superscript A	upper I superscript B	lower i
Carbohydrate	A, triangle	B, circle	none

Figure a is a table for the 3 alleles for the A B O blood groups and their carbohydrates. Each allele codes for an enzyme that may add a specific carbohydrate (designated by the superscript on the allele and shown as a triangle or circle) to red blood cells.

The table is as follows:

Figure b is of blood group genotypes and phenotypes. There are six possible genotypes, resulting in four different phenotypes. The table is as follows:

Genotype	upper I superscript A, upper I superscript A, or upper I superscript A, lower i	upper I superscript B, upper I superscript B, or upper I superscript B, lower i	upper I superscript A, upper I superscript B	lower i, lower i

Red blood cell with surface carbohydrates	red blood cell with triangles on its surface	red blood cell with circles on its surface	red blood cell with triangles and circles	red blood cell with nothing on its surface
Phenotype (blood group)	A	B	A B	o

VISUAL SKILLS *Based on the surface carbohydrate phenotype in part (b), what are the dominance relationships among the alleles?*

*For suggested answer, see*Appendix A.**

Pleiotropy

So far, we have treated Mendelian inheritance as though each gene affects only one phenotypic character. Most genes, however, have multiple phenotypic effects, a property called **pleiotropy** (from the Greek *pleion*, more). In humans, for example, pleiotropic alleles are responsible for the multiple symptoms associated with certain hereditary diseases, such as cystic fibrosis and sickle-cell disease, discussed later in this chapter. In the garden pea, the gene that determines flower color also affects the color of the coating on the outer surface of the seed, which can be gray or white. Given the intricate molecular and cellular interactions responsible for an organism's development and physiology, it isn't surprising that a single gene can affect a number of characters.

Extending Mendelian Genetics for Two or More Genes

Dominance relationships, multiple alleles, and pleiotropy all have to do with the effects of the alleles of a single gene. We now consider two situations in which two or more genes are involved in determining a particular phenotype. In the first case, called epistasis, one gene affects the phenotype of another because the two gene products interact; in the second case, called polygenic inheritance, multiple genes independently affect a single trait.

Epistasis

In **epistasis** (from the Greek for "standing upon"), the phenotypic expression of a gene at one locus alters that of a gene at a second locus. An example will help clarify this concept. In Labrador retrievers (commonly called "Labs"), black coat color is dominant to brown. Let's designate B and b as the two alleles for this character. For a Lab to have brown fur, its genotype must be bb; these dogs are called chocolate Labs. But there is more to the story. A second gene determines whether or not pigment will be deposited in the hair.

The dominant allele, symbolized by E, results in the deposition of either black or brown pigment, depending on the genotype at the first locus. But if the Lab is homozygous recessive for the second locus ee, then the coat is yellow, regardless of the genotype at the black/brown locus (yellow Labs). In this case, the gene for pigment deposition E/e is said to be *epistatic to* the gene that codes for black or brown pigment B/b.

What happens if we mate black Labs that are heterozygous for both genes BbEe? Although the two genes affect the same phenotypic character (coat color), they follow the law of independent assortment. Thus, our breeding experiment represents an F1 dihybrid cross, like those that produced a 9:3:3:1 ratio in Mendel's experiments. We can use a Punnett square to represent the genotypes of the F2 offspring **(Figure 14.12)**. As a result of epistasis, the phenotypic ratio among the F2 offspring is 9 black to 3 chocolate to 4 yellow Labs. Other types of epistatic interactions produce different ratios, but all are modified versions of 9:3:3:1.

Figure 14.12

```
An example of epistasis.
```

Black labs (uppercase B lowercase b uppercase E lowercase e) mated; results shown in a Punnett square with 16 boxes. 9 black, 3 chocolate and 4 golden labs result.

Figure b is of blood group genotypes and phenotypes. There are six possible genotypes, resulting in four different phenotypes. The table is as follows:

Genotype	upper I superscript A, upper I superscript A, or upper I superscript A, lower i	upper I superscript B, upper I superscript B, or upper I superscript B, lower i	upper I superscript A, upper I superscript B	lower i, lower i
Red blood cell with surface carbohydrates	red blood cell with triangles on its surface	red blood cell with circles on its surface	red blood cell with triangles and circles	red blood cell with nothing on its surface
Phenotype (blood group)	A	B	A B	O

This Punnett square illustrates the genotypes and phenotypes predicted for offspring of matings between two black Labrador retrievers of genotype BbEe. The E/e gene, which is epistatic to the B/b gene coding for hair pigment, controls whether or not pigment of any color will be deposited in the hair.

VISUAL SKILLS *Compare the four squares in the lower right of this Punnett square with those in*Figure 14.8*.* Explain the genetic basis for the difference between the ratio*(9:3:4)*of phenotypes seen in this cross and the*9:3:3:1*ratio seen in*Figure 14.8*.*

For suggested answer, see Appendix A.

Polygenic Inheritance

Mendel studied characters that could be classified on an either-or basis, such as purple versus white flower color. But many characters, such as human skin color and height, are not one of two discrete characters, but instead vary in the population in gradations along a continuum. These are called **quantitative characters**. Quantitative variation usually indicates **polygenic inheritance**, an additive effect of two or more genes on a single phenotypic character. (In a way, this is the converse of pleiotropy, where a single gene affects several phenotypic characters.) Height is a good example of polygenic inheritance: In 2014, a genomic study of over 250,000 people found almost 700 genetic variations in over 180 genes that affect height. Many variations were in or near genes involved in biochemical pathways affecting growth of the skeleton, but others were associated with genes not obviously related to growth. Another study in 2018, of 300,000 individuals, identified 124 genes that affect hair color. Although simplifying can sometimes be valuable in understanding genetic principles, as we'll see shortly, the classic idea of a single gene determining eye color or hair color or other such traits is an oversimplification. (Earlobe attachment, an example used until a decade ago in this textbook, was shown in 2017 to be affected by nearly 50 genes!)

Skin pigmentation in humans is also controlled by many separately inherited genes—378 at latest count, many of which are involved in the production of melanin skin pigments. Here, we'll simplify the story in order to understand the concept of polygenic inheritance. Let's consider three genes, with a dark-skin allele for each gene (A, B, or C) contributing one "unit" of darkness (also a simplification) to the phenotype and being incompletely dominant to the other, light-skin allele (a, b, or c). In our model, an AABBCC person would be very dark, whereas an aabbcc individual would be very light. An AaBbCc person would have skin of an intermediate shade. Because the alleles have a cumulative effect, the genotypes AaBbCc and AABbcc would make the same genetic contribution (three units) to skin darkness. There are seven skin color phenotypes that could result from a mating between AaBbCc heterozygotes, as shown in **Figure 14.13**. In a large number of such matings, the majority of offspring would be expected to have intermediate phenotypes (skin color in the middle range). You can graph the predictions from the Punnett square in the **Scientific Skills Exercise**. Environmental factors, such as exposure to the sun, also affect the skin color phenotype.

Figure 14.13

```
A simplified model for polygenic inheritance of skin color.
```

Possible combinations of alleles (genes A, B, C) for skin color between parents heterozygous for all three genes. Skin colors range across seven phenotypes from very light to very dark.

Parental genotypes for two black labs: both uppercase B lowercase b uppercase E lowercase e.

Gametes	one-fourth uppercase B and E	one-fourth lowercase b uppercase E	one-fourth Uppercase B lowercase e	one-fourth lowercase b and e
one-fourth uppercase B and E	Uppercase B, B, E, E (black lab)	Uppercase B, lowercase b, Uppercase E, E (black lab)	Uppercase B, B, E, lowercase e (black lab)	Uppercase B, lowercase b, Uppercase E, lowercase e (black lab)
one-fourth lowercase b uppercase E	Uppercase B, lowercase b, upper case E, E (black lab)	lowercase b, b, Uppercase E, E (brown lab)	Uppercase B, lowercase b, uppercase E, lowercase e (black lab)	lowercase b, b, Uppercase E, lowercase e (brown lab)
one-fourth Uppercase B lowercase e	Uppercase B, B, E, lowercase e (black lab)	Uppercase B, lowercase b, Uppercase E, lowercase e (black lab)	Uppercase B, B, lowercase e, e (golden lab)	Uppercase B, lowercase b, lowercase e, e (golden lab)
one-fourth lowercase b and e	Uppercase B, lower case b, uppercase E, lowercase e (black lab)	lowercase b, b, Uppercase E, lowercase e (brown lab)	Uppercase B, lowercase b, e, e (golden lab)	lowercase b, b, e, e (golden lab)

Gametes	one eighth a b c all lowercase	one eighth Uppercase A, lowercase b c	one eighth Lowercase a uppercase B	one eighth Lowercase a lowercase b	one eighth Uppercase A uppercase B	one eighth Uppercase A lowercase b

			lowercase c	uppercase C	lowercase c	uppercase C
one eighth a b c all lowercase	a a b b c c all lowercase	Uppercase A, a b b cc all lowercase	Lowercase a a, uppercase B lowercase b, cc both lowercase	a a b b all lowercase, uppercase C lowercase c	Uppercase A lowercase a uppercase B, b c c all lowercase	Uppercase A, a b b all lowercase, uppercase C lowercase c
one eighth uppercase A lowercase b c	Uppercase A, a b b c c all lowercase	AA both uppercase, b b c c all lowercase	Uppercase A lowercase a uppercase B, bcc all lowercase	Uppercase A, a b b all lowercase, uppercase C lowercase c	AAB all uppercase, bcc all lowercase	AA both uppercase, b b both lowercase, uppercase C lowercase c
one eighth lowercase a uppercase B lowercase c	aa both uppercase, uppercase B, bcc all lowercase	Uppercase A lowercase a uppercase B, bcc all lowercase	aa both lowercase, BB both uppercase, cc both lowercase	aa both lowercase, uppercase B lowercase b uppercase C lowercase c	Uppercase A lowercase a, BB both uppercase, cc both lowercase	Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c
one eighth ab both lowercase, uppercase C	a a b b all lowercase, uppercase C lowercase c	Uppercase A, a b b all lowercase, uppercase C lowercase c	aa both lowercase, uppercase B lowercase b uppercase C lowercase c	a a b b all lowercase, C C both uppercase	Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c	Uppercase A, a b b all lowercase, C C both uppercase
one eighth A B both uppercase, lowercase c	Uppercase A lowercase a uppercase B, b c c all lowercase	A A B all uppercase, bcc all lowercase	Uppercase A lowercase a, B B both uppercase, c c both lowercase	Uppercase A lowercase a uppercase B lowercase b uppercase	A A B B all uppercase, cc both lowercase	A A B all uppercase, lowercase b uppercase C lowercase c

				C lowercase c			
one eighth uppercase A lowercase b uppercase C	Uppercase A, a b b all lowercase, uppercase C lowercase c	AA both uppercase, bb both lowercase, uppercase C lowercase c	Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c	Uppercase A, a b b all lowercase, C C both uppercase	A A B all uppercase, lowercase b uppercase C lowercase c	AA both uppercase, bb both lowercase, C C both uppercase	
one eighth lowercase a uppercase BC	aa both lowercase, uppercase B lowercase b uppercase C lowercase c	Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c	aa both lowercase, uppercase B lowercase b uppercase C lowercase c	aa both lowercase uppercase B lowercase b, CC both uppercase	Uppercase A lowercase a, BBC all uppercase, lowercase c	Uppercase A lowercase a uppercase B lowercase b, CC both uppercase	
one eighth A B C all uppercase	Uppercase A lowercase a uppercase B lowercase b uppercase C lowercase c	A A B all uppercase, lowercase b uppercase C lowercase c	Uppercase A lowercase a, B B C all uppercase, lowercase c	Uppercase A lowercase a uppercase B lowercase b, C C both uppercase	A A B B C all uppercase, lowercase c	A A B all uppercase, lowercase b, C C both uppercase	

There are 7 skin phenotypes shown, with increasing darkness in skin color related to the number of dark-skin alleles present (0 to 6).

one sixty-fourth phenotypes are very light with no dark-skin alleles.

six sixty fourths phenotypes are light with one dark-skin allele.

fifteen sixty-fourths phenotypes are slightly darker, with 2 dark-skin alleles.

twenty sixty fourths phenotypes are slightly darker, with 3 dark-skin alleles.

fifteen sixty thirds phenotypes are brown with 4 dark-skin alleles.

six sixty fourths phenotypes are darker brown with 5 dark skin alleles.

one sixty fourth phenotypes are dark brown with 6 dark-skin alleles.

In this model, three separately inherited genes affect skin color. (The reported number is actually 378 genes.) The heterozygous individuals (AaBbCc) represented by the two rectangles at the top of this figure each carry three dark-skin alleles (black circles, which represent A, B, or C) and three light-skin alleles (white circles, which represent a, b, or c). The Punnett square shows all possible genetic combinations in gametes and offspring of hypothetical matings between these two. The results are summarized by phenotypic frequencies (fractions) under the Punnett square. (The phenotypic ratio is 1:6:15:20:15:6:1.)

HHMI BioInteractive Video

The Biology of Skin Color

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Scientific Skills Exercise

Making a Histogram and Analyzing a Distribution Pattern

AP Science Practices 2, 4

What Is the Distribution of Phenotypes Among Offspring of Two Parents Who Are Both Heterozygous for Three Additive Genes? Human skin color is a polygenic trait that is determined by the additive effects of many different genes. In this exercise, you will work with a simplified model of skin color genetics where only three genes are assumed to affect the darkness of skin color and where each gene has two alleles—dark or light (see Figure 14.13). In this model, each dark allele contributes equally to the darkness of skin color, and each pair of alleles segregates independently of any other pair. Using a type of graph called a histogram, you will determine the distribution of phenotypes of offspring with different numbers of dark-skin alleles. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)

A group of people with different colored hair, eyes, and skin.

How This Model Is Analyzed To predict the phenotypes of the offspring of parents heterozygous for the three genes in our simplified model, we can use the Punnett square in Figure 14.13. The heterozygous individuals (AaBbCc) represented by the two rectangles at the top of that figure each carry three dark-skin alleles (black circles, which represent A, B, or C) and three light-skin alleles (white circles, which represent a, b, or c). The Punnett square shows all the possible genetic combinations in gametes and in offspring of a large number of hypothetical matings between these heterozygotes.

Predictions from the Punnett Square If we assume that each square in the Punnett square represents one offspring of the heterozygous AaBbCc parents, then the squares below show the possible skin color phenotypes and their predicted frequencies. Below the squares is the number of dark-skin alleles for each phenotype.

A table lists the phenotypes and the number of dark-skin alleles.

The table is as follows:

1 over 64	6 over 64	15 over 64	20 over 64	15 over 64	6 over 64	1 over 64
0	1	2	3	4	5	6

Parents both normal phenotypes, and both with genotypes uppercase A lowercase a.

Probabilities for phenotypes in their children are one-fourth normal, one-half normal carriers, and one-fourth albino.

Gametes	Uppercase A	Lowercase a
Uppercase A	A A both uppercase Phenotype: normal	Uppercase A lower case a Phenotype: Normal (carrier)
Lowercase a	Uppercase A lower case a Phenotype: Normal (carrier)	a a both lowercase Phenotype: albino

The square is as follows:

Gametes	Uppercase D	Lowercase d
Lowercase d	Uppercase D lowercase d Phenotype: Dwarf	d d both lowercase Phenotype: Normal
Lowercase d	Uppercase D lowercase d Phenotype: Dwarf	d d both lowercase Phenotype: Normal

Further Reading R. A. Sturm, A golden age of human pigmentation genetics, *Trends in Genetics* 22:464–468 (2006).

INTERPRET THE DATA

- A histogram is a bar graph that shows the distribution of numeric data (here, the number of dark-skin alleles). To make a histogram of the allele distribution, put skin color (as the number of dark-skin alleles) along the x-axis and predicted number of offspring (out of 64) with each phenotype on the y-axis. There are no gaps in these allele data, so draw the bars next to each other with no space in between.
- You can see that the skin-color phenotypes are not distributed uniformly.
- Which phenotype has the highest frequency? Draw a vertical dashed line through that bar.
- Distributions of values like this one tend to show one of several common patterns. Sketch a rough curve that approximates the values and look at its shape. Is it symmetrically distributed around a central peak value (a "normal distribution," sometimes called a bell curve); is it skewed to one end of the x-axis or the other (a "skewed distribution"); or does it show two apparent groups of frequencies (a "bimodal distribution")? Explain the reason for the curve's shape. (It will help to read the text description that supports Figure 14.13.)

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Nature and Nurture: The Environmental Impact on Phenotype

Another departure from simple Mendelian genetics arises when the phenotype for a character depends on environment as well as genotype. A single tree, locked into its inherited genotype, has leaves that vary in size, shape, and greenness, depending on their exposure to wind and sun. For humans, nutrition influences height, exercise alters build, sun-tanning darkens the skin, and experience improves performance on intelligence tests. Even identical twins, who are genetic equals, accumulate phenotypic differences as a result of their unique experiences.

Whether human characters are more influenced by genes or the environment—in everyday terms, nature versus nurture—is a debate that we will not attempt to settle here. We can say, however, that a genotype generally is not associated with a rigidly defined phenotype, but rather with a range of phenotypic possibilities due to environmental influences **(Figure 14.14)**. For some characters, such as the ABO blood group system, the phenotypic range is extremely narrow; that is, a given genotype mandates a very specific phenotype. Other characters, such as a person's blood count of red and white cells, vary quite a bit,

depending on such factors as the altitude, the customary level of physical activity, and the presence of infectious agents.

Figure 14.14

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The effect of environment on phenotype.
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Two photos. Hydrangeas grown in basic soil are bright pink in figure a. Hydrangeas grown in acidic soil with free aluminum are bluish-violet in figure b.

The outcome of a genotype lies within a phenotypic range that depends on the environment in which the genotype is expressed. For example, the acidity and free aluminum content of the soil affect the color of hydrangea flowers, which range from pink (basic soil) to blue-violet (acidic soil). Free aluminum is necessary for bluer colors.

Generally, the phenotypic range is broadest for polygenic characters. Environment contributes to the quantitative nature of these characters, as we have seen in the continuous variation of skin color. Geneticists refer to such characters as **multifactorial**, meaning that many factors, both genetic and environmental, collectively influence phenotype.

BBC Video

Genetics vs. Environment

image

Play

image

Play

A Mendelian View of Heredity and Variation

We have now broadened our view of Mendelian inheritance by exploring degrees of dominance as well as multiple alleles, pleiotropy, epistasis, polygenic inheritance, and the phenotypic impact of the environment. How can we integrate these refinements into a comprehensive theory of Mendelian genetics? The key is to make the transition from the reductionist emphasis on single genes and phenotypic characters to the emergent properties of the organism as a whole, one of the themes of this book.

The term *phenotype* can refer not only to specific characters, such as flower color and blood group, but also to an organism in its entirety—*all* aspects of its physical appearance, internal anatomy, physiology, and behavior. Similarly, the term *genotype* can refer to an organism's entire genetic makeup, not just its alleles for a single genetic locus. In most cases, a gene's impact on phenotype is affected by other genes and by the environment. In this integrated view of heredity and variation, an organism's phenotype reflects its overall genotype and unique environmental history.

Considering all that can occur in the pathway from genotype to phenotype, it is indeed impressive that Mendel could uncover the fundamental principles governing the transmission of individual genes from parents to offspring. Mendel's laws of segregation and of independent assortment explain heritable variations in terms of alternative forms of genes (hereditary "particles," now known as the alleles of genes) that are passed along, generation after generation, according to simple rules of probability. This theory of inheritance is equally valid for peas, flies, fishes, birds, and human beings—indeed, for any organism with a sexual life cycle. Furthermore, by extending the principles of segregation and independent assortment to help explain such hereditary patterns as epistasis and quantitative characters, we begin to see how broadly Mendelian genetics applies. From Mendel's abbey garden came a theory of inheritance that anchors modern genetics. In the last section of this chapter, we will apply Mendelian genetics to human inheritance, with emphasis on the transmission of hereditary diseases.